

EFGP-SING M-step: collapsed GP-block update for kernel hyperparameters

Big picture

In the E-step notes, we rewrote the SING-GP drift update as an EFGP pseudo-regression problem:

$$q(f) \text{ update} = \text{EFGP regression on uncertain latent velocities.}$$

The M-step asks: given the current inferred latent posterior $q(x)$, how do we update the model parameters?

There are two relevant model-parameter blocks:

1. **GP drift kernel hyperparameters**, such as lengthscale and output scale.
2. **Non-GP model parameters**, such as emissions C, d, R , diffusion Σ , and possible input effects B .

The EFGP-specific question is the first one:

Can we update GP kernel hyperparameters without losing the $O(N) + f(M)$ structure?

The answer is yes.

The recommended update is a **collapsed GP-block M-step**:

Fix $q(x)$, and for each θ re-optimize $q(f) = q_{\theta}^*(f)$.

This is not strict variational EM. It is a collapsed/profile update of the GP drift block.

1. What SING does in the M-step

1.1 SING's general M-step

In the SING paper, the learned model parameters are collectively denoted

$$\Theta = \{\theta, C, d\}.$$

Here θ denotes prior drift parameters, and C, d are output/emission parameters.

For drift estimation, SING alternates between inference and learning. During learning, they optimize drift parameters θ using Adam on the approximate ELBO, while output parameters C, d are updated in closed form for Gaussian observations.

So the SING pattern is:

E-step: update $q(x)$ and, for GP drift, $q(f)$.

M-step: update drift hyperparameters and possibly emissions.

For SING-GP with an RBF kernel, the paper says the only learnable GP prior drift parameters are the RBF lengthscale and output scale. The inducing-point posterior is updated during the E-step, not the M-step.

1.2 What we replace

For GP drift, SING uses sparse inducing points and learns kernel hyperparameters by Adam/backprop through the ELBO, while using closed-form posterior drift updates for $q(u)$ in the E-step.

We replace this with:

Use EFGP pseudo-regression and compute kernel gradients directly in feature space.

So the replacement is not in the emission update. It is specifically in the GP drift hyperparameter update.

SING-GP M-step object:

Adam/backprop update for sparse-GP kernel hyperparameters.

EFGP-SING M-step object:

direct feature-space gradient for collapsed pseudo-GP regression objective.

2. Reminder: the E-step pseudo-regression problem

From the E-step, with $q(x)$ fixed, the GP drift update can be viewed as heteroskedastic GP regression on latent velocities.

For drift coordinate r ,

$$y_{i,r}^{\text{pseudo}} = \frac{x_{i+1,r} - x_{i,r}}{\Delta_i}.$$

If x_i were known, the pseudo-regression model would be

$$y_{i,r}^{\text{pseudo}} = f_r(x_i) + \epsilon_{i,r},$$

with

$$\epsilon_{i,r} \sim N\left(0, \frac{\sigma_r^2}{\Delta_i}\right).$$

Equivalently, the precision weight is

$$\lambda_i = \frac{\Delta_i}{\sigma_r^2}.$$

With uncertain latents, the exact SING expression is written using expectations and Stein corrections. But computationally, after sampling marginal latent locations or transition pairs, we get an ordinary weighted pseudo-regression problem.

Let X be the sampled latent input cloud. Let W_r be the diagonal matrix of pseudo-observation weights. Let y_r be the vector of pseudo-responses or Stein-corrected pseudo-response weights.

In EFGP,

$$f_r(x) = \phi_\theta(x)^* w_r,$$

with

$$\phi_\theta(x) = D_\theta F_x,$$

and

$$w_r \sim N(0, I).$$

Define

$$\Phi_\theta = F_X D_\theta.$$

Then the E-step posterior over Fourier weights is

$$q(w_r) = N(\mu_r, Q_r),$$

where

$$Q_r^{-1} = A_{\theta,r} = I + \Phi_\theta^* W_r \Phi_\theta,$$

and

$$A_{\theta,r} \mu_r = h_{\theta,r},$$

with

$$h_{\theta,r} = \Phi_{\theta}^* W_r y_r.$$

In the feature-space notation,

$$A_{\theta,r} = I + D_{\theta} F_X^* W_r F_X D_{\theta},$$

and

$$h_{\theta,r} = D_{\theta} F_X^* W_r y_r.$$

For the Stein-corrected marginal-sampling estimator, $h_{\theta,r}$ is a sum of the increment term and derivative correction, but the structure is the same: cached NUFFT summaries multiplied by D_{θ} and frequency multipliers.

3. Strict variational EM versus collapsed GP-block update

3.1 Strict variational EM

Strict variational EM says:

Fix $q(x)$ and $q(f)$, update θ .

After the E-step at current hyperparameters θ_0 , we have

$$q(w_r) = N(\mu_r, Q_r).$$

Strict VEM holds μ_r and Q_r fixed while updating θ .

The fixed- q objective for the pseudo-regression block is

$$\mathcal{L}_r^{\text{fixed-}q}(\theta) = \mathbb{E}_{q(w_r)}[\log p_{\theta}(y_r \mid X, w_r)] + \text{terms independent of } \theta.$$

In feature-space form, this is

$$\mathcal{L}_r^{\text{fixed-}q}(\theta) = h_{\theta,r}^* \mu_r - \frac{1}{2} \text{tr} [A_{\theta,r} (Q_r + \mu_r \mu_r^*)] + \text{const.}$$

The gradient is

$$\partial_{\theta} \mathcal{L}_r^{\text{fixed-}q} = \text{Re} [(\partial_{\theta} h_{\theta,r})^* \mu_r] - \frac{1}{2} \mu_r^* (\partial_{\theta} A_{\theta,r}) \mu_r - \frac{1}{2} \text{tr} [Q_r \partial_{\theta} A_{\theta,r}].$$

Here μ_r and Q_r are fixed from the E-step.

This is the closest analogue of SING's variational EM framing.

3.2 Collapsed GP-block update

The collapsed view says:

$$\boxed{\text{Fix } q(x), \text{ but for each } \theta \text{ set } q(f) = q_\theta^*(f).}$$

That is,

$$q_\theta^*(w_r) = \arg \max_{q(w_r)} \mathcal{L}(q(x), q(w_r), \theta).$$

For the Gaussian pseudo-regression block,

$$q_\theta^*(w_r) = N(\mu_{\theta,r}, Q_{\theta,r}),$$

where

$$Q_{\theta,r} = A_{\theta,r}^{-1},$$

and

$$\mu_{\theta,r} = A_{\theta,r}^{-1} h_{\theta,r}.$$

Then the collapsed/profile objective is

$$\mathcal{L}_r^{\text{coll}}(\theta) = \max_{q(w_r)} \mathcal{L}(q(x), q(w_r), \theta).$$

For the Gaussian pseudo-regression block, up to constants independent of θ ,

$$\mathcal{L}_r^{\text{coll}}(\theta) = \frac{1}{2} h_{\theta,r}^* A_{\theta,r}^{-1} h_{\theta,r} - \frac{1}{2} \log |A_{\theta,r}|.$$

The gradient is

$$\partial_\theta \mathcal{L}_r^{\text{coll}} = \text{Re} [(\partial_\theta h_{\theta,r})^* \mu_{\theta,r}] - \frac{1}{2} \mu_{\theta,r}^* (\partial_\theta A_{\theta,r}) \mu_{\theta,r} - \frac{1}{2} \text{tr} [A_{\theta,r}^{-1} \partial_\theta A_{\theta,r}].$$

This is the version we recommend implementing.

It is not strict variational EM because $q(w_r)$ changes as θ changes during the M-step. It is a collapsed/profile variational update for the GP drift block.

3.3 When are strict VEM and collapsed gradients equivalent?

They are equivalent at the current E-step optimum.

Suppose the E-step was run at θ_0 , so

$$Q_r = A_{\theta_0, r}^{-1},$$

and

$$\mu_r = A_{\theta_0, r}^{-1} h_{\theta_0, r}.$$

Then at $\theta = \theta_0$,

$$\partial_\theta \mathcal{L}_r^{\text{fixed-}q} = \partial_\theta \mathcal{L}_r^{\text{coll}}.$$

This is the envelope theorem.

But after a finite step away from θ_0 , if $q(w_r)$ is not recomputed, then generally

$$Q_r \neq A_{\theta, r}^{-1},$$

and

$$\mu_r \neq A_{\theta, r}^{-1} h_{\theta, r}.$$

So the two views are no longer equivalent.

3.4 Why the collapsed view is cleaner for whitened EFGP

In the whitened EFGP representation,

$$f_r(x) = \phi_\theta(x)^* w_r,$$

and

$$w_r \sim N(0, I).$$

The hyperparameters θ enter through

$$\phi_\theta(x) = D_\theta F_x.$$

Changing θ changes the map from weights to functions:

$$w_r \mapsto f_r.$$

So holding $q(w_r)$ fixed while changing D_θ is not exactly the same as holding the function-space distribution $q(f_r)$ fixed.

This makes strict VEM slightly awkward in whitened coordinates.

The collapsed view avoids this issue by saying: for each θ , define the corresponding optimal posterior

$$q_{\theta}^*(w_r) = N(A_{\theta,r}^{-1}h_{\theta,r}, A_{\theta,r}^{-1}).$$

Then optimize the resulting profile objective.

Important wording:

θ remains a model/kernel hyperparameter.

But

$q^*(f)$ depends on θ .

Do not say that θ is a parameter of $q(f)$. Say instead:

We optimize a collapsed GP drift objective with $q(f)$ profiled out.

4. What exactly replaces SING's M-step for GP hyperparameters?

4.1 SING-GP version

For GP drift, SING uses sparse inducing points and learns kernel hyperparameters with gradient-based optimization of the approximate ELBO. In the RBF GP-drift example, the only learned GP kernel parameters are the RBF lengthscale and output scale. The inducing posterior update is done in the E-step.

So schematically, SING-GP does:

1. Fix current $q(x)$.
2. Use sparse-GP closed-form updates for $q(u)$ during the E-step.
3. In the M-step, optimize GP kernel hyperparameters θ with Adam/backprop on the ELBO.

4.2 EFGP replacement

We replace the sparse-GP M-step gradient with a direct EFGP feature-space gradient of the collapsed pseudo-regression objective.

For each drift coordinate r :

$$A_{\theta,r} = I + D_{\theta}T_rD_{\theta},$$

where

$$T_r = F_X^* W_r F_X.$$

And

$$h_{\theta,r} = D_\theta z_r,$$

where

$$z_r = F_X^* W_r y_r.$$

For Stein-corrected marginal sampling, z_r is a sum of cached NUFFT outputs:

$$z_r = z_{1,r} + z_{2,r},$$

with

$$z_{1,r} = F_X^* a_r,$$

and

$$z_{2,r} = \sum_j (2\pi i \xi_j) \odot F_X^* c_{j,r}.$$

Depending on how we place D_θ , this can be written as

$$h_{\theta,r} = D_\theta z_{1,r} + \sum_j (2\pi i \xi_j) \odot D_\theta F_X^* c_{j,r}.$$

Equivalently, the key property is:

$h_{\theta,r}$ is built from cached NUFFT summaries plus diagonal spectral weights.

Then the collapsed M-step objective is

$$\mathcal{L}^{\text{coll}}(\theta) = \sum_r \left[\frac{1}{2} h_{\theta,r}^* A_{\theta,r}^{-1} h_{\theta,r} - \frac{1}{2} \log |A_{\theta,r}| \right] + \text{const.}$$

This replaces SING's backprop-through-sparse-GP ELBO update for kernel hyperparameters.

5. Direct gradient for the collapsed EFGP M-step

5.1 Derivative of the precision

Let

$$A_{\theta,r} = I + D_\theta T_r D_\theta.$$

Assume the Fourier grid and sampled latent cloud X are fixed during this M-step block, so T_r is fixed.

Then

$$\partial_{\theta_p} A_{\theta,r} = (\partial_{\theta_p} D_\theta) T_r D_\theta + D_\theta T_r (\partial_{\theta_p} D_\theta).$$

A matvec with $\partial_{\theta_p} A_{\theta,r}$ is:

1. multiply by $\partial_{\theta_p} D_\theta$;
2. apply the BTTB Toeplitz operator T_r ;
3. multiply by D_θ ;
4. add the symmetric term.

Therefore,

$$v \mapsto (\partial_{\theta_p} A_{\theta,r})v$$

costs

$$O(M \log M).$$

No $O(NM)$ operation appears.

5.2 Derivative of the RHS

If

$$h_{\theta,r} = D_\theta z_r,$$

and z_r is cached, then

$$\partial_{\theta_p} h_{\theta,r} = (\partial_{\theta_p} D_\theta) z_r.$$

This is a diagonal multiply.

For Stein-corrected marginal sampling, all the $F_X^* a_r$ and $F_X^* c_{j,r}$ summaries are cached. The derivatives still only hit D_θ , and possibly spectral frequency factors if the Fourier grid itself changes.

With a fixed Fourier grid, this costs

$$O(M).$$

5.3 Gradient formula

For each kernel hyperparameter θ_p ,

$$\partial_{\theta_p} \mathcal{L}_r^{\text{coll}} = \text{Re} [(\partial_{\theta_p} h_{\theta,r})^* \mu_{\theta,r}] - \frac{1}{2} \mu_{\theta,r}^* (\partial_{\theta_p} A_{\theta,r}) \mu_{\theta,r} - \frac{1}{2} \text{tr} [A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}],$$

where

$$\mu_{\theta,r} = A_{\theta,r}^{-1} h_{\theta,r}.$$

The first two terms are deterministic once $\mu_{\theta,r}$ is solved.

The trace term is estimated stochastically.

6. Hutchinson trace estimator for the M-step

The hard term is

$$\text{tr} [A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}].$$

Use Rademacher probes

$$v_j \in \{-1, +1\}^M.$$

Then

$$\text{tr} [A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}] \approx \frac{1}{J} \sum_{j=1}^J v_j^* A_{\theta,r}^{-1} (\partial_{\theta_p} A_{\theta,r}) v_j.$$

For each probe:

1. Compute

$$b_{j,p} = (\partial_{\theta_p} A_{\theta,r}) v_j.$$

2. Solve

$$A_{\theta,r} u_{j,p} = b_{j,p}$$

using CG.

3. Accumulate

$$v_j^* u_{j,p}.$$

Each CG iteration uses the EFGP precision matvec

$$v \mapsto A_{\theta,r}v = v + D_{\theta}T_rD_{\theta}v.$$

So the trace estimator costs

$$O(JkM \log M)$$

per output dimension and per block of hyperparameters, up to batching.

6.1 Batched trace estimator

For multiple kernel hyperparameters $\theta_1, \dots, \theta_P$, build a batch of right-hand sides

$$b_{j,p} = (\partial_{\theta_p} A_{\theta,r})v_j.$$

Then solve all systems

$$A_{\theta,r}u_{j,p} = b_{j,p}$$

with batched CG.

This is the same computational pattern as EFGP hyperparameter learning:

kernel gradients reduce to BTTB matvecs, CG solves, and Hutchinson traces.

7. Fixed-grid caching and amortization

At fixed $q(x)$, fixed sampled pseudo-inputs X , fixed weights W_r , and fixed Fourier grid, cache:

$$T_r = F_X^* W_r F_X$$

as a BTTB generator.

Also cache RHS summaries:

$$z_r = F_X^* W_r y_r,$$

or for Stein-corrected marginal sampling,

$$z_{1,r} = F_X^* a_r,$$

and

$$z_{2,j,r} = F_X^* c_{j,r}.$$

Then during M-step gradient updates, changing θ changes only

$$D_{\theta}$$

and

$$\partial_{\theta_p} D_{\theta}.$$

Thus, after setup, inner M-step gradient calculations have no $O(N)$ work except if we refresh samples or change the Fourier grid.

This mirrors the standard EFGP hyperparameter-learning amortization: for a fixed Fourier grid, $O(N)$ summaries can be cached, and additional gradient steps mostly cost only M -space CG and Toeplitz work.

8. What if the Fourier grid depends on θ ?

This is the main caveat.

If changing the lengthscale also changes the Fourier frequencies ξ_k , then

$$F_X$$

changes, so both

$$T_r = F_X^* W_r F_X$$

and

$$z_r = F_X^* W_r y_r$$

must be rebuilt.

That costs

$$O(N + M \log M).$$

Still not $O(NM)$, but it removes the amortized fixed-grid advantage.

The practical strategy is:

Hold the Fourier grid fixed for several M-step gradient steps, update D_{θ} , then refresh the grid occasionally.
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This is exactly the same strategy as standard EFGP hyperparameter learning.

9. How this differs from strict VEM in practice

9.1 Strict VEM version

Strict VEM would do:

1. Run E-step at θ_0 .
2. Store

$$q(w_r) = N(\mu_r, Q_r).$$

3. In the M-step, hold μ_r, Q_r fixed.
4. Optimize

$$\mathcal{L}^{\text{fixed-}q}(\theta).$$

This is variational EM in the strict sense.

But in whitened EFGP coordinates, holding $q(w_r)$ fixed while changing D_θ is philosophically awkward because the function distribution changes when the map $w_r \mapsto f_r$ changes.

9.2 Collapsed version

The collapsed update does:

1. Fix $q(x)$.
2. For each θ , solve

$$A_{\theta,r} \mu_{\theta,r} = h_{\theta,r}.$$

3. Use

$$Q_{\theta,r} = A_{\theta,r}^{-1}$$

in the trace term.

4. Optimize

$$\mathcal{L}^{\text{coll}}(\theta).$$

This is not strict VEM, because $q(w_r)$ changes during the hyperparameter update.

But it is a clean profile update:

$$\mathcal{L}^{\text{coll}}(\theta) = \max_{q(f)} \mathcal{L}(q(x), q(f), \theta).$$

For a methods section, describe this honestly as:

With $q(x)$ fixed, we perform a collapsed update of the GP drift block,

optimizing the ELBO with $q(f)$ available in closed form for each θ .

10. Other M-step parameters: what does SING update?

10.1 Drift parameters

SING updates drift parameters θ using Adam on the approximate ELBO.

For non-GP drift classes, θ can mean neural-network weights or polynomial coefficients.

For GP drift, θ means kernel hyperparameters. In the RBF GP-drift experiment, these are the lengthscale and output scale.

In the neural-data SING-GP application, they use a more structured smoothly switching linear kernel, whose kernel hyperparameters are optimized with Adam during variational EM.

Our replacement only applies to GP kernel hyperparameters. It does not apply to neural drift weights or polynomial drift coefficients, because those are not EFGP kernel parameters.

10.2 Emission parameters C, d

SING also updates output parameters C, d in some experiments.

For Gaussian emissions,

$$y_i \sim N(Cx_i + d, R),$$

the update for C, d can be done in closed form as expected least squares under $q(x)$.

That part is independent of EFGP.

So we should do one of two things:

1. **Initially hold C, d fixed**, as SING does in some neural-data GP experiments.
2. **Later add the same closed-form expected least-squares update** used by SING.

The EFGP kernel M-step does not interfere with the emission update.

10.3 Observation noise R

If we learn Gaussian observation noise R , it can also be updated from expected residuals:

$$y_i - Cx_i - d.$$

This uses moments under $q(x_i)$:

$$\mathbb{E}[x_i], \quad \mathbb{E}[x_i x_i^\top].$$

No EFGP machinery is needed.

If not central, hold R fixed initially.

10.4 Diffusion covariance Σ

The diffusion covariance Σ appears in the Euler transition noise:

$$x_{i+1} \mid x_i, f \sim N(x_i + \Delta_i f(x_i), \Delta_i \Sigma).$$

Updating Σ would require expected transition residuals:

$$x_{i+1} - x_i - \Delta_i f(x_i).$$

Given $q(x)$ and $q(f)$, this uses the same drift moments from the E-step:

$$\mathbb{E}[f(x_i)],$$

and

$$\mathbb{E}[f(x_i) f(x_i)^\top].$$

This is feasible with EFGP posterior predictive moments, but it is an additional moving piece.

Recommendation:

Hold Σ fixed until the GP kernel M-step works.

10.5 Input effects B

If the model includes inputs,

$$x_{i+1} - x_i \approx \Delta_i f(x_i) + \Delta_i B v_i + \epsilon_i,$$

then B can be updated like a weighted regression problem once $q(x)$ and $q(f)$ are fixed.

The pseudo residual is

$$x_{i+1} - x_i - \Delta_i f(x_i),$$

and the predictor is

$$\Delta_i v_i.$$

This again uses $q(x)$ moments and GP drift posterior moments, but it is not an EFGP bottleneck.

Recommendation:

Hold B fixed initially, then add a separate weighted regression update if needed.

11. Recommended staged implementation

Stage 1: E-step only

Hold all model parameters fixed:

$$\theta, \Sigma, C, d, R, B.$$

Implement:

$$q(x) \leftrightarrow q(f)$$

with the EFGP drift block.

Goal:

Verify EFGP E-step scaling and correctness.

Stage 2: collapsed M-step for GP kernel hyperparameters only

Fix

$$C, d, R, \Sigma, B.$$

Update only GP kernel hyperparameters:

$$\theta_{\text{kernel}}.$$

Use the collapsed objective:

$$\mathcal{L}^{\text{coll}}(\theta) = \sum_r \left[\frac{1}{2} h_{\theta,r}^* A_{\theta,r}^{-1} h_{\theta,r} - \frac{1}{2} \log |A_{\theta,r}| \right].$$

Compute gradients directly with:

$$\partial_{\theta_p} h_{\theta,r}, \quad \partial_{\theta_p} A_{\theta,r}, \quad A_{\theta,r}^{-1} v, \quad \text{tr}[A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}].$$

Goal:

Replace SING-GP's Adam/backprop kernel update with EFGP direct gradients.

Stage 3: add emission updates

Add closed-form Gaussian emission updates:

$$C, d = \arg \min_{C, d} \sum_i \mathbb{E}_{q(x_i)} [\|y_i - Cx_i - d\|_{R^{-1}}^2].$$

Optionally update R .

Goal:

Match SING's treatment of output parameters.

Stage 4: optionally add Σ and B

Once the GP drift and emissions are stable, optionally update:

$$\Sigma$$

and

$$B.$$

These are separate moment/regression updates, not part of the EFGP kernel bottleneck.

12. Algorithm block: collapsed EFGP M-step

Given:

- current $q(x)$,
- current sampled pseudo-inputs X ,
- pseudo-responses or Stein-corrected RHS summaries,
- fixed Fourier grid $\{\xi_k\}_{k=1}^M$,
- current kernel hyperparameters θ ,
- diagonal spectral matrix D_θ ,
- drift dimensions $r = 1, \dots, D_{\text{out}}$,
- Hutchinson probes J ,
- CG tolerance.

Goal: update GP kernel hyperparameters θ using a collapsed GP-block objective.

1. Cache EFGP summaries.

For each drift coordinate r , build or reuse

$$T_r = F_X^* W_r F_X.$$

Store T_r as a BTTB/Toeplitz generator.

Also cache

$$z_r = F_X^* W_r y_r,$$

or the Stein-corrected components

$$z_{1,r} = F_X^* a_r,$$

and

$$z_{2,j,r} = F_X^* c_{j,r}.$$

2. Build current spectral weights.

From θ , compute

$$D_\theta.$$

For each hyperparameter θ_p , compute

$$\partial_{\theta_p} D_\theta.$$

3. Define the precision operator.

For each r , define

$$A_{\theta,r} v = v + D_\theta T_r D_\theta v.$$

4. Build the RHS.

For pair-sampled pseudo-regression,

$$h_{\theta,r} = D_\theta z_r.$$

For Stein-corrected marginal sampling,

$$h_{\theta,r} = D_\theta z_{1,r} + \sum_j (2\pi i \xi_j) \odot D_\theta z_{2,j,r}.$$

5. Solve for the collapsed posterior mean.

Use CG:

$$A_{\theta,r}\mu_{\theta,r} = h_{\theta,r}.$$

6. Compute deterministic gradient terms.

For each θ_p , compute

$$\partial_{\theta_p} h_{\theta,r}.$$

Compute

$$(\partial_{\theta_p} A_{\theta,r})\mu_{\theta,r}.$$

Accumulate

$$\text{Re} [(\partial_{\theta_p} h_{\theta,r})^* \mu_{\theta,r}] - \frac{1}{2} \mu_{\theta,r}^* (\partial_{\theta_p} A_{\theta,r}) \mu_{\theta,r}.$$

7. Estimate trace terms.

Draw Rademacher probes

$$v_j \in \{-1, +1\}^M.$$

For each p, j , compute

$$b_{p,j} = (\partial_{\theta_p} A_{\theta,r}) v_j.$$

Solve

$$A_{\theta,r} u_{p,j} = b_{p,j}.$$

Estimate

$$\text{tr}[A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}] \approx \frac{1}{J} \sum_j v_j^* u_{p,j}.$$

8. Assemble gradient.

For each θ_p ,

$$\partial_{\theta_p} \mathcal{L}^{\text{coll}} = \sum_r \left\{ \text{Re} [(\partial_{\theta_p} h_{\theta,r})^* \mu_{\theta,r}] - \frac{1}{2} \mu_{\theta,r}^* (\partial_{\theta_p} A_{\theta,r}) \mu_{\theta,r} - \frac{1}{2} \text{tr}[A_{\theta,r}^{-1} \partial_{\theta_p} A_{\theta,r}] \right\}.$$

9. Take a gradient step.

Update θ using Adam or another optimizer.

10. Refresh grid or samples if needed.

If the Fourier grid is held fixed, repeat steps 2--9 cheaply.

If the Fourier grid changes, rebuild

$$T_r$$

and

$$z_r.$$

13. Complexity

Let:

- N be the number of pseudo-inputs,
- $M = m^d$ be the number of Fourier features,
- P be the number of kernel hyperparameters,
- J be the number of Hutchinson probes,
- k be the number of CG iterations,
- D_{out} be the number of drift output dimensions.

Setup per fixed grid:

$$O(D_{\text{out}}N + D_{\text{out}}M \log M).$$

Collapsed posterior mean solves:

$$O(D_{\text{out}}kM \log M).$$

Deterministic gradient terms:

$$O(PD_{\text{out}}M \log M).$$

Trace terms:

$$O(PJD_{\text{out}}kM \log M).$$

The key point is:

No $O(NM)$ feature matrices and no dense $M \times M$ matrices.

With fixed Fourier grid and cached summaries, repeated M-step gradient updates are mostly M -space operations.

14. Final recommendation

Implement the M-step in this order:

1. **Collapsed GP kernel update only.**

Update only kernel hyperparameters:

$$\theta_{\text{kernel}}.$$

Use the collapsed objective:

$$\mathcal{L}^{\text{coll}}(\theta) = \sum_r \left[\frac{1}{2} h_{\theta,r}^* A_{\theta,r}^{-1} h_{\theta,r} - \frac{1}{2} \log |A_{\theta,r}| \right].$$

2. **Keep emissions fixed initially.**

This matches the simpler SING-GP neural-data setting where C, d are initialized and held fixed.

3. **Add closed-form C, d updates later.**

This matches the synthetic drift-estimation setting.

4. **Hold Σ fixed initially.**

Updating Σ is feasible but requires additional drift posterior moment bookkeeping.

5. **Describe the method honestly.**

Do not call the kernel update strict VEM if using the collapsed objective.

Say:

With $q(x)$ fixed, we optimize a collapsed GP drift objective,

profiling out $q(f)$ through its closed-form EFGP posterior.

The main conceptual summary is:

SING learns GP kernel hyperparameters by backprop through the sparse-GP ELBO.

EFGP-SING replaces this with direct feature-space gradients of a collapsed pseudo-GP regression obje

And the main computational summary is:

The M-step is feasible with the same EFGP tools: BTTB matvecs, CG, Hutchinson, and cached NUFF